Question 2 - CI/CD workflow report

Question 2: Prepare a CI/CD workflow report describing the pipeline architecture, stages, automation results, and observations with supporting screenshots.

Repository: https://github.com/ksamirdev/ensp461-devops-lab1

Objective

Build and deploy a small Flask text statistics service through a Git and Jenkins workflow, with every change tested before release.

Tools and environment

Component	Used
Host	Ubuntu 24.04 LTS (Docker, systemd)
Source control	Git, GitHub (public repository)
CI/CD	Jenkins 2.580.2 LTS, OpenJDK 21
Application	Python, Flask 3.1.0, pytest 8.3.4
Runtime	systemd service on port 5000

The lab host is an Ubuntu 24.04 Docker container booted with systemd, so Jenkins and the application run as real services. The container holds no GitHub credentials, so pushes were sent from the host machine using its existing GitHub login, while Jenkins pulls from the public repository.

Pipeline architecture

```
git push
  -> GitHub main
  -> Jenkins pollSCM (H/2 * * * *)
  -> Checkout -> Build -> Test (JUnit results) -> Package (archived tarball)
  -> Deploy (/opt/text-stats, systemd restart, /healthz check)
```

Stages

- Checkout: Jenkins pulls the Jenkinsfile and source from GitHub.
- Build: creates a venv and installs dependencies.
- Test: runs 5 pytest cases and publishes results.
- Package: archives text-stats-BUILD.tar.gz.
- Deploy: installs the archive, restarts the service and checks health.

Automation results

Build	Cause	Result	Duration	Deployed
1	Manual (Build Now equivalent)	SUCCESS	15 s	1.0.0
2	SCM change	SUCCESS	7 s	1.1.0

Evidence

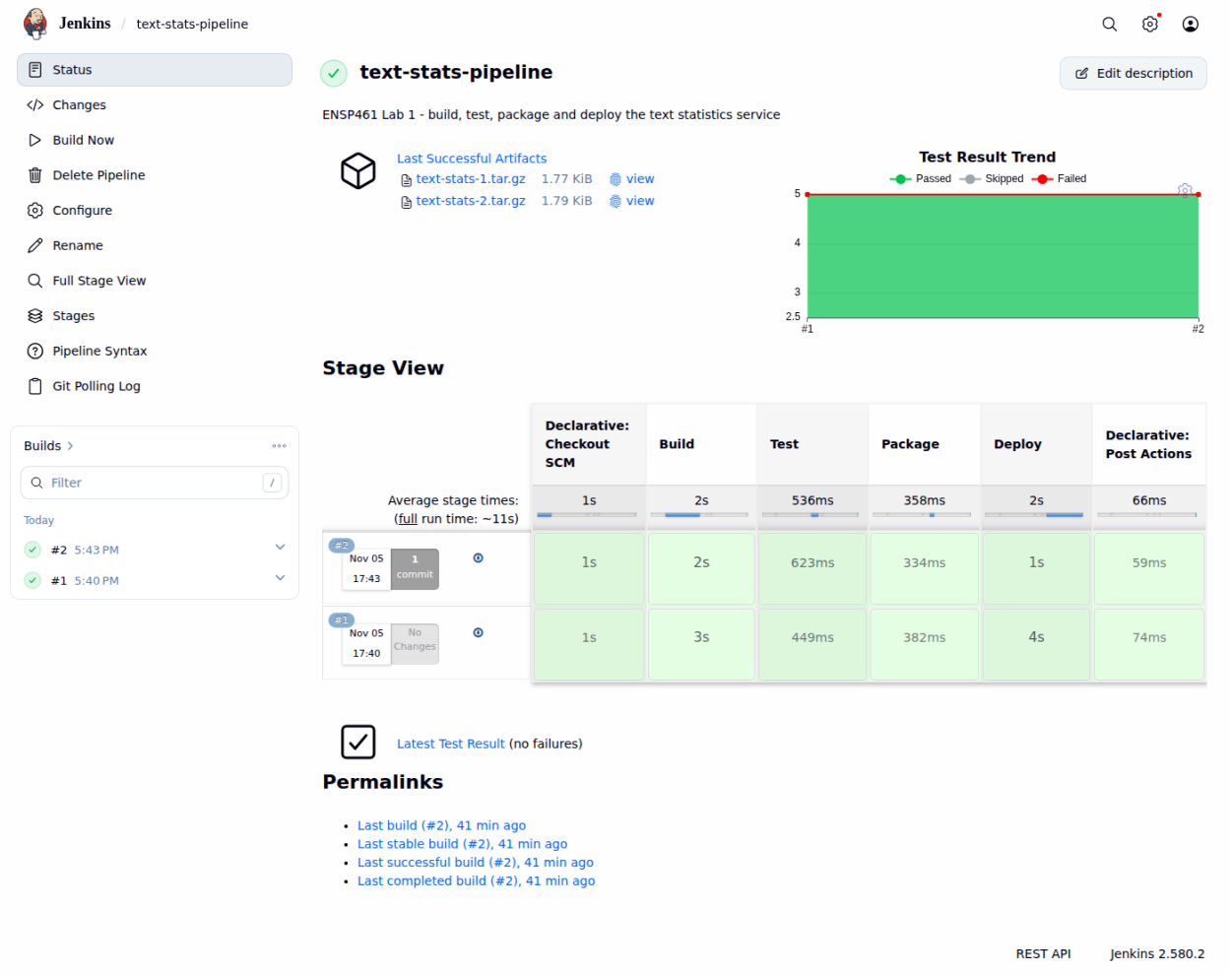


Fig: Stage view

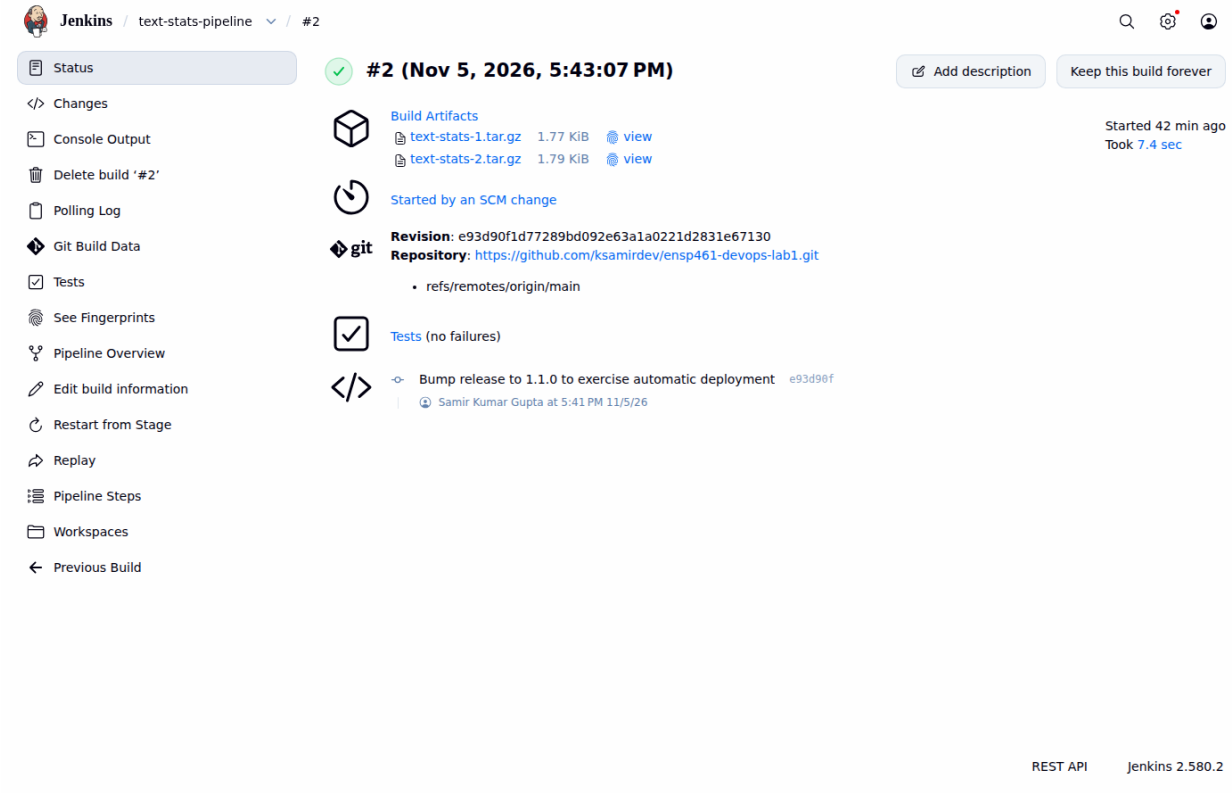


Fig: Automatic build

Observations

- Build 1 was started manually and finished with SUCCESS in about 15 seconds.
- Pipeline script from SCM means the pipeline definition is versioned with the code it builds.
- Using a public HTTPS remote means Jenkins needs no stored Git credentials.
- Build 2 was started by an SCM change, without clicking Build Now, and finished in about 7 seconds.
- After build 2 the /healthz endpoint reported release 1.1.0, confirming the new code went live.

Conclusion

A push to main now reaches the running service without manual steps, and a failing test or health check stops the pipeline with a visible failure.